Topic 6C: Alkenes
18. know the general formula for alkenes
19. know that alkenes and cycloalkenes are unsaturated hydrocarbons
20. understand the bonding in alkenes in terms of σ- and π- bonds
21. know what is meant by the term ‘electrophile’
22. understand the addition reactions of alkenes with: i hydrogen, in the presence of a nickel catalyst, to form an alkane *Knowledge of the application of this reaction to the manufacture of margarine by catalytic hydrogenation of unsaturated vegetable oils is expected*. ii halogens to produce dihalogenoalkanes iii hydrogen halides to produce halogenoalkanes iv steam, in the presence of an acid catalyst, to produce alcohols v potassium manganate(VII), in acid conditions, to oxidise the double bond and produce a diol
23. understand that heterolytic bond fission of a covalent bond results in the formation of ions

Students should:
24. understand the mechanism of the electrophilic addition reactions between alkenes and: i halogens ii hydrogen halides, including addition to unsymmetrical alkenes iii other given binary compounds *Use of the curly arrow notation is expected − curly arrows should start from either a bond or from a lone pair of electrons*. *Knowledge of the relative stability of primary, secondary and tertiary carbocation intermediates is expected*.
25. know the qualitative test for a C=C double bond using bromine or bromine water
26. know that alkenes form polymers through addition polymerisation *Be able to identify the repeat unit of an addition polymer given the monomer, and vice versa.*
27. know that waste polymers can be separated into specific types of polymer for: i recycling ii incineration to release energy iii use as a feedstock for cracking
28. understand, in terms of the use of energy and resources over the life cycle of polymer products, that chemists can contribute to the more sustainable use of materials
29. understand how chemists limit the problems caused by polymer disposal by: i developing biodegradable polymers ii removing toxic waste gases caused by incineration of plastics